

Vivek Vishwakarma

PhD Candidate (Thesis submitted) · Department of Civil Engineering

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Research areas of Interest

Structural durability · Corrosion · Fatigue · Non-Destructive Testing (NDT) · Fracture Mechanics · Acoustic Emission · Acoustic Signal Processing

Education

Year	Degree	Institution	CGPA / %
2026	Ph.D. (Thesis submitted)	Indian Institute of Technology Roorkee	8.143
2013	Postgraduate (M.Tech.)	Indian Institute of Technology Roorkee	6.840
2011	Graduate (B.E.)	Shri G.S. Institute of Tech. and Science	71.19%
2006	Intermediate (Class XII)	Modi Public School, Kota	68.00%
2004	Matriculate (Class X)	Kendriya Vidyalaya, Damoh	66.00%

Professional Experience

Assistant Professor | *Jabalpur Engineering College* Jan 2018 – Jul 2021

- Taught core Civil Engineering subjects including Structural Analysis, RCC Design, and Engineering Mechanics to undergraduate students across multiple semesters.
- Supervised undergraduate final-year projects on structural assessment and construction materials, mentoring students through experimental and analytical work.

Assistant Professor | *The ICFAI University* Jul 2014 – Jan 2018

- Taught core Civil Engineering subjects including Structural Analysis, RCC Design, and Engineering Mechanics to undergraduate students across multiple semesters.
- Supervised undergraduate final-year projects on structural assessment and construction materials, mentoring students through experimental and analytical work.

Assistant Project Manager | *Jones Lang LaSalle* Jul 2013 – Jun 2014

- Managed civil construction of a 2B+G+7 storey hostel building in Mumbai, coordinating contractors, consultants, and clients while monitoring timelines, costs, and quality compliance across structural and finishing works.
- Conducted site inspections and prepared technical progress reports ensuring adherence to IS code specifications and contractual milestones.

Research Publications (Q1-SCIE Indexed, Peer-reviewed)

1. Vishwakarma, V., & Ray, S. (2026). A hybrid analytical framework for coupled corrosion-fatigue in RC beams: From anodic flaw evolution to global stiffness decay. *International Journal of Fatigue*, 213, 109847. <https://doi.org/10.1016/j.ijfatigue.2026.109847>
2. Vishwakarma, V., & Ray, S. (2025b). Coupled corrosion and fatigue in reinforced concrete beams: Experimental interpretation of mechanical and electrochemical synergy. *International Journal of Fatigue*, 203, 109318. <https://doi.org/10.1016/j.ijfatigue.2025.109318>
3. Vishwakarma, V., & Ray, S. (2025a). Probability-based prediction of crack propagation in lightly reinforced concrete beams using acoustic emission technique. *Applied Acoustics*, 241, 111035. <https://doi.org/10.1016/j.apacoust.2025.111035>
4. Masih, V. W., Ruiz, G., Maganty, S. T., Sharma, A., Ray, S., Vishwakarma, V., De La Rosa, Á., & Yu, R. C. (2026). Modified Brazilian Test for studying crack initiation and fracture behavior in SFRC using DIC & AE. *Construction and Building Materials*, 523, 146241. <https://doi.org/10.1016/j.conbuildmat.2026.146241>

Research Publications (International conferences)

1. Vivek Vishwakarma and Sonalisa Ray, “Behaviour of reinforced cement concrete under coupled corrosion fatigue,” in *FraMCoS XII*, IA-FraMCoS, **2025**.
2. Vivek Vishwakarma and Sonalisa Ray, “Prediction of crack path in reinforced concrete using acoustic emission analysis,” *Procedia Structural Integrity*, Elsevier, **2024**.
3. Vivek Vishwakarma and Sonalisa Ray, “Effects of coupled corrosion and fatigue on the performance of reinforced cement concrete,” in *FraMCoS-XI*, IA-FraMCoS, **2023**.

Research & Consultancy Projects

Fracture performance of RC members in coupled corrosion and fatigue environment | *SERB, India* Aug 2021 – Jun 2024

- Research focused on fracture and fatigue performance of reinforced concrete in a coupled corrosion-fatigue environment.

Nonlinear FE analysis of concrete damage due to impact loading | *NPIU (TEQIP-III)* Sep 2019 – Sep 2020

- Collaborated with NPIU (TEQIP-III) on funded research involving nonlinear FE analysis of concrete under impact loading using ABAQUS.

Structural Design of First Floor Slab and Beam — S.O.I Office Building, Vijay Nagar Jabalpur | *CPWD* Nov 2020 – Jul 2021

- Structural design of first floor level slab and beam as per relevant IS codes.

Structural Soundness Evaluation — S.O.I Office Building, Vijay Nagar Jabalpur | *CPWD* Sep 2020 – Nov 2020

- Consultancy project for evaluation of structural soundness of existing construction.

Structural Stability Testing — Building No. 1, Grey Iron Foundry, Jabalpur | *Grey Iron Foundry* Jun 2021 – Jul 2021

- Structural audit of industrial steel shed.

Inspection & Recommendation — Netaji Subhash Chandra Bose Medical College, Jabalpur | *PWDPIU, MP* Aug 2019 – Sep 2019

- Conditional structural survey of Existing Spinal Injury Centre for assessment of feasibility of additional construction over the existing structure.

Inspection & Structural Audit — High Court Buildings | *Hon'ble High Court of Madhya Pradesh* Jun 2019 – Jul 2019

- Structural audit of buildings in the High Court complex, including a heritage building.

Awards & Achievements

- **Best Paper Award** — CARRS 2025: 2nd International Conference on Condition Assessment, Rehabilitation & Retrofitting of Structures.
- **MHRD Scholarship** for Post-graduation through GATE.

Skills

Software MATLAB, COMSOL Multiphysics, ABAQUS, STAAD Pro

Languages English (Speaking, Reading, Writing), Hindi (Speaking, Reading, Writing)

References

Dr. Sonalisa Ray

Professor

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